

ZEYU (ALBAN) LI

AI Integration Engineer | Applied Scientist | Full-Stack AI Developer

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SUMMARY

Cornell Ph.D. in Biological & Environmental Engineering with a **Computer Science** background, specializing in context engineering, LLM evaluation, and building production AI systems for complex real-world workflows. Over 5 years of experience spanning **environmental science, biomedical engineering, architecture, materials science, and nucleic acid engineering** -- achieving 78% first-match precision in a production semantic search system, winning Best Hardware Prize at Cornell's AI Health Hackathon, and engineering an open-source agent harness cataloging 33 systematic LLM failure patterns. Shipped 4 production AI applications (multilingual healthcare voice pipeline with clinical decision support, researcher matching platform, agentic desktop assistant, consumer trust-and-safety app with content moderation), engineered end-to-end ETL pipelines for a DoD-funded 11 sq. km field campaign, and published an open-source LLM reliability framework validated across 130+ sessions. Author of 4 publications (3 published, 1 under review at *Nature*) and 2 granted patents.

SKILLS

AI & LLM Integration: LLM API integration (Claude, Gemini); context engineering; agentic workflow design & multi-tool orchestration; RAG pipelines (FAISS, BM25, Reciprocal Rank Fusion); function calling; streaming ASR; LLM evaluation & guardrails; hallucination detection; content moderation & sentiment analysis; human-in-the-loop verification; AI-assisted development (Copilot, Cursor, Claude Code)

Data Engineering & Pipelines: Python (Pandas, NumPy, Seaborn, GeoPandas); ETL & automated multi-source data pipelines; data validation & quality assurance; geospatial analytics; time-series synchronization; schema enforcement & data governance

Full-Stack & Deployment: React 18/19; Electron; Firebase Auth & Realtime DB; Vercel; Streamlit; WebSocket & Web Audio API (AudioWorklet); SMS/notification APIs; REST APIs; web scraping & indexing

ML & Modeling: TensorFlow/Keras; PyTorch; XGBoost; semantic embeddings (SPECTER2); hybrid retrieval systems; image classification

Scientific Instrumentation & Lab: qPCR; DNA/RNA extraction & purification; SEM/EDS; rheometry; DLS; fluorometry (Qubit); spectrophotometry (NanoDrop); bio-cleanroom; microfluidics; 3D printing (DIW, SLA, FDM)

Languages: Python | Java | JavaScript (React) | English (fluent) | Mandarin (native) | Cantonese (proficient)

AI PRODUCTS & INTEGRATION PROJECTS

Smart Translator (Kinapse) -- Agentic macOS Desktop AI Assistant

Co-built (2-person team) | Python, Electron, Gemini API, Firebase, PyObjC, WebSocket

- **Co-developed an agentic macOS desktop app with parallel multi-tool orchestration:** real-time streaming ASR (Gemini Live API, transparent 50-second session rotation with zero frame loss), 7-category intent classifier with dual-signal task switching (app-window hint at zero latency + transcript contradiction detection), and a multimodal context pipeline integrating screenshot vision analysis, bilingual Chinese-English NLP, and active-window detection -- targeting sub-1-second end-to-end latency.
- **Designed a two-layer RAG memory system** combining structured user profiles with episodic interaction history indexed via 3072-dimensional Gemini embeddings and cosine similarity retrieval, enabling per-user prompt personalization. Built a behavior analytics dashboard with composite scoring across 6 evaluation dimensions (EMA-smoothed) to track user skill development over time.

ER Companion -- Healthcare AI Voice Pipeline & Wearable Integration

Cornell AI Health Hackathon | Best Hardware Prize | React 19, Gemini Live API, WebSocket, Firebase, Wearable Hardware

- **Problem addressed:** 155M U.S. ED visits/year; non-English-speaking patients experience 5.4 min longer to be roomed and 15.6 min longer for disposition -- a direct result of real-time communication gaps between patients and clinical staff.
- **Built a continuous-listening, multilingual voice pipeline** (Gemini 2.5 Flash native audio, WebSocket bidirectional streaming, AudioWorklet low-latency PCM processing) across 8 languages with automatic language detection and real-time switching; structured clinical data extraction via 4 function-call types following the OPQRST triage framework; dual-layer clinical documentation preserving patient's exact words alongside AI-generated clinical paraphrases for nurse dashboard review.

- **Implemented dual-track red flag detection:** 150+ multilingual regex patterns (cardiac, respiratory, neurological, psychological) for instant frontend alerts, complemented by Gemini function-call clinical severity assessment with ESI-2 threshold scoring and special-population adjustments (elderly, pediatric, immunocompromised); periodic automated check-ins every 7 minutes with pain-score delta tracking. | GitHub: github.com/AlbanLi0314/er-companion

Resonance -- AI-Powered Researcher Matching Platform

Google DeepMind Gemini Hackathon | Deployed on Streamlit | RAG, FAISS, BM25, Gemini API

- **Designed and deployed a hybrid semantic search system** indexing 230 Cornell researchers: SPECTER2 citation-trained embeddings (768-dim) with FAISS dense retrieval, combined with BM25 sparse information retrieval via Reciprocal Rank Fusion reranking, achieving 77.8% P@1, 0.884 nDCG@5, and 0.847 MRR on 45 human-labeled test queries.
- **Integrated Gemini API for multi-turn conversational search:** paper analysis, interest discovery through structured dialogue, query expansion (+4.5% precision uplift), and natural-language match explanation -- enabling researchers to find collaborators through conversation rather than keyword search. | Live: [resonance-aounkapvlodwxrz2n49we.streamlit.app](https://aounkapvlodwxrz2n49we.streamlit.app)

Echo -- Full-Stack AI Consumer App

Production | Vercel | React 18, Firebase Auth, Gemini API, NVC Analysis

- **Independently designed and shipped a production consumer web app end-to-end:** double-blind matching with mutual-intent confirmation via Firebase real-time sync, SMS verification (Firebase Auth + reCAPTCHA v3), bilingual EN/Chinese interface, and graceful API degradation ensuring uninterrupted service.
- **Integrated Gemini API for real-time content moderation and user safety:** NVC-based sentiment analysis detecting 6 expression-trap categories (judgment, blame, hidden demands, rhetorical questions), keyword-pattern vulnerability detection (50+ bilingual crisis signals) with automatic message blocking and 988 crisis resource routing for at-risk users. Live: echo-app-ebon.vercel.app

Thesis Harness -- Open-Source LLM Reliability Framework

Open-Source | Claude API, Python, Context Engineering, LLM Evaluation | 130+ Sessions, 11 Protocols, 33 Error Patterns

- **Engineered an open-source agent harness for reliable LLM-assisted technical writing:** 11 modular protocols implementing context engineering (session state management, five-layer knowledge base), systematic guardrails (mandatory pre-flight checklists, error prevention rules), and multi-stage human-in-the-loop verification pipelines -- validated across 130+ sessions producing a doctoral thesis. | GitHub: github.com/AlbanLi0314/thesis-harness
- **Built an LLM evaluation framework cataloging 33 systematic failure modes** (hallucination, arithmetic errors, context drift, passive-context decay) across 12 categories, with pattern-specific prevention rules that reduced error recurrence to near-zero; includes a dual-AI cross-verification pipeline for hallucination detection in academic citations.

Automated Field Campaign Data Pipeline -- DoD-Funded Environmental Intelligence

Python, Pandas, GeoPandas, Seaborn | 33-hour field deployment, 11 sq. km lake, 278 sampling stations

- **Engineered a multi-source ETL pipeline** for a DoD SERDP-funded 33-hour field campaign: ingested approximately 1,000 qPCR measurements from 278 geographically distributed sampling stations and approximately 1,400 GPS drifter coordinates, applied automated data validation and quality checks, temporal synchronization, and coordinate standardization, outputting structured multi-table datasets for hydrodynamic model benchmarking.
- **Built a modular geospatial visualization framework** (approximately 20 Python scripts) with strict schema enforcement and data governance enabling reproducible, zero-touch regeneration of publication-ready geospatial maps and statistical plots directly from raw data. Engineered procedural 3D visualization workflows in Blender using Geometry Nodes and Python scripting, replacing manual illustration with code-driven pipelines for manuscript figures.

RESEARCH EXPERIENCE

Graduate Research Assistant

Feb 2021 - Present

DNA Materials Lab, Advisor: Prof. Dan Luo | Cornell University, Ithaca, NY

Environmental Science | Hydrology | Field Instrumentation

DNA-Barcoded PLGA Microparticle Tracers -- DoD-Funded Lake-Scale Field Deployment

- **Led design, fabrication, and deployment of synthetic eDNA tracers** (DNA-barcoded PLGA microspheres) across 11 sq. km of Cayuga Lake, coordinating 4 disciplines and 7 research groups; conducted rigorous qPCR detection in a positive-pressure bio-cleanroom, achieving detection 7 km downstream from less than 1 mg DNA release.
- **First-author, Environmental Science & Technology (IF = 11.3), 2025.** DOI: 10.1021/acs.est.5c11071. Supported by DoD SERDP. Seminars at ACS Fall 2023 (San Francisco), UIUC, Penn State, and Cornell.

Architecture | Materials Science | Computational Design

Self-Healing DNA-Al³⁺ Hydrogel Composites for Structural Repair (PolyTile 4.0)

- **Developed 3D-printable biomass-DNA hydrogel platform** with three independent crosslinking modes (thermal, UV-reversible psoralen, ionic Al³⁺), enabling time-programmed shape locking and repeatable self-healing across 50+ formulations; integrated as embedded adhesive in 3D-printed vascular ceramic tiles for autonomous crack repair.
- **Co-first-author, ACADIA 2024** (premier international conference in computational design & digital fabrication). DOI: 10.52842/conf.acadia.2024.1.089. Supported by Cornell SPROUT Awards.

Biochemical Engineering | Nucleic Acid Technology | Process Engineering

Espresso Machine-Inspired Nucleic Acid Extraction Platform

- **Invented a pod-based continuous-flow nucleic acid extraction system** using a regenerable perfluorocarbon capture medium; achieved approximately 200 mg DNA per pod and over 90% capture efficiency across 0.01-4 M salinity, reducing consumables cost by approximately 91% vs. conventional ethanol precipitation while yielding enzyme-compatible, sequencing-ready outputs.

Bioengineering | Microfluidics | Nucleic Acid Engineering

NucleoMesh -- Continuous Nucleic Acid Production Platform (Nature, Under Review)

- **Co-designed and fabricated microfluidic chip reactors** for perpetual enzymatic nucleic acid amplification in a closed-loop system; performed all materials characterization (SEM, rheology) and developed a bead-based RNA purification method achieving up to 94% recovery.
- **2nd author -- Wang, D., Li, Z., et al., "A Chip Reactor for Perpetual Nucleic Acid Production and On-chip Information Processing."** Nature, under review (2026).

Undergraduate Research Assistant & Senior Research Assistant

Jun 2017 - Jan 2021

Microfabrication & Surface Materials Lab, Supervisor: Prof. Kangning Ren | Hong Kong Baptist University

Materials Science | Surface Engineering | Anti-Icing

Durable Superhydrophobic Polymer Surfaces for Icing Control

- **Engineered all-perfluoropolymer superhydrophobic surfaces** via thermal pressing with master molds from photolithography, two-photon lithography, and etching; validated durability under abrasion (1,500 cycles, contact angle at least 150 degrees), UV, hydraulic pressure, and 100,000+ folds. Led experimental study demonstrating approximately 27% reduction in droplet contact time at minus 10 degrees Celsius using electrostatic pre-charging.
- **3rd-author, The Innovation (Cell Press, IF = 33.2), 2023.** Co-inventor, US Patent 11,839,998 (granted 2023). Best Undergraduate Thesis Award, HKBU 2019.

Research Exchange Trainee

Jun 2018 - Aug 2018

Molecular Basis of Disease Program | Georgia State University, Atlanta, GA

- **Characterized transcription factor-DNA binding kinetics (PU.1)** via LSPR assays, quantifying sequence-specific binding affinities in real time. Best Poster Presentation Award.

TEACHING & AI CURRICULUM

Co-Instructor & Teaching Assistant | Cornell University, BEE Department

Feb 2021 - Present

- **BEE 3400 -- Design and Analysis of Biomaterials (Co-Instructor, ~50% of content, approximately 45 students):** Designed and delivered a 6-lecture AI Tooling Module covering LLM fundamentals, prompt & context engineering, agentic workflow design, and AI-assisted data processing. Organized and led a 40-student Biomaterials Hackathon with structured deliverables and oral defense format, recruited and managed student assistant team.
- **Guest lecturer on AI-assisted engineering workflows** for 2 additional undergraduate courses; topics included transformer attention, context engineering, the 40-20-40 rule, and practical agent design patterns.
- **Teaching assistant across 7 courses:** BEE 7600 (Nucleic Acid Engineering), BEE 5600/3600 (Molecular & Cellular Bioengineering), BEE 3400, BEE 2600, ENGRI 1337, ENGRG 5330 (Engineering Professionalism), BEE 5730/4730 (Watershed Engineering).
- **Directly supervised 3 undergraduate researchers** on end-to-end wet-lab protocols and data analysis workflows.

EDUCATION

Cornell University

Ithaca, NY

Biological and Environmental Engineering | Concentration: Materials Science, Biological Engineering, Bioenvironmental Engineering

Ph.D. -- DNA Materials Lab, Advisor: Prof. Dan Luo | Expected May 2026

M.S. | Aug 2024

M.Eng. | May 2020

Hong Kong Baptist University

Kowloon, Hong Kong

Chemistry (Major) | Computer Science (Minor)

B.Sc. (Hons) -- Microfabrication & Surface Materials Lab, Supervisor: Prof. Kangning Ren | Nov 2019

PUBLICATIONS

Wang, D., **Li, Z.**, Li, J., Han, Y., Sun, T., Li, F., Liu, P. "A Chip Reactor for Perpetual Nucleic Acid Production and On-chip Information Processing." Nature, under review (2026).

Li, Z., Ramón, C. L., Koeberle, A., et al. "Tracing Environmental DNA Transport in Large Lakes with Synthetic DNA Microparticles and Hydrodynamic Modeling." Environmental Science & Technology (IF = 11.3), 2025. DOI: 10.1021/acs.est.5c11071

He, C. *, **Li, Z. ***, Wang, L. X., et al. "PolyTile 4.0: Self-healing Ceramic Tiles." ACADIA 2024. DOI: 10.52842/conf.acadia.2024.1.089

Li, W., Chan, C. W., **Li, Z.**, et al. "All-perfluoropolymer, nonlinear stability-assisted monolithic surface combines topology-specific superwettability with ultradurability." The Innovation (IF = 33.2), 4(2), 100389, 2023. DOI: 10.1016/j.xinn.2023.100389

PATENTS

Ren, K., Wu, H., Wang, Z., Yao, S., Ong, B., Li, W., **Li, Z.**, Sun, H., & Chan, C.W. "Crack engineering as a new route for the construction of arbitrary hierarchical architectures." US Patent 11,839,998 (granted 2023).

Li, Q., **Li, Z.**, & Lin, Z. "Reactor and method of spiral propulsion biomass continuous thermal cracking." Chinese Patent 201711214139.6 (granted 2017).